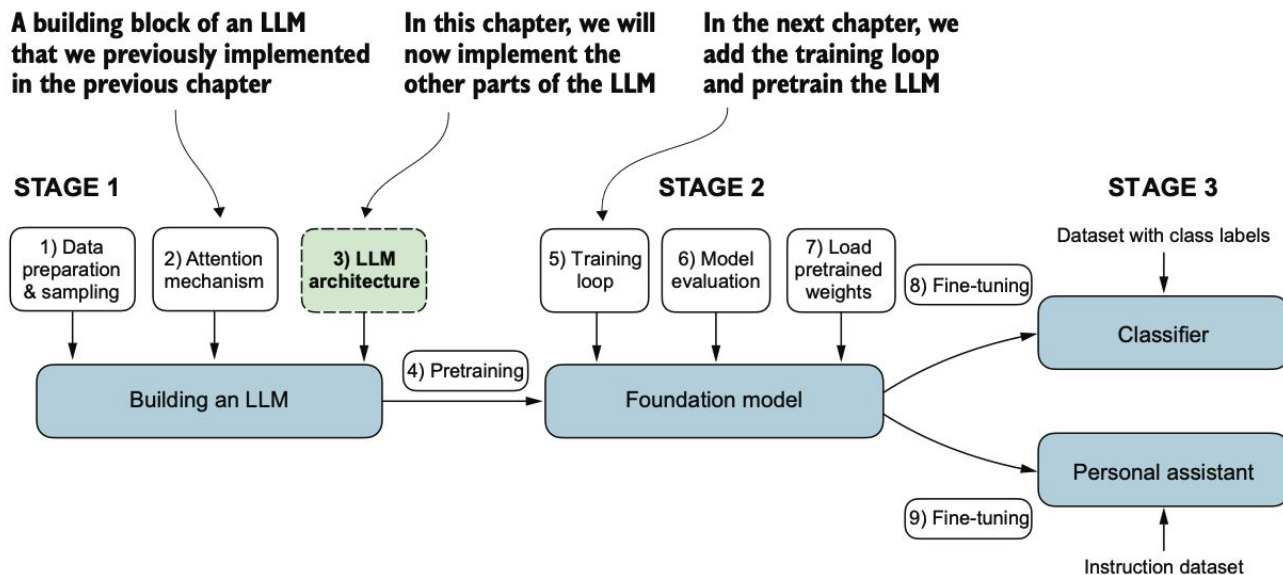


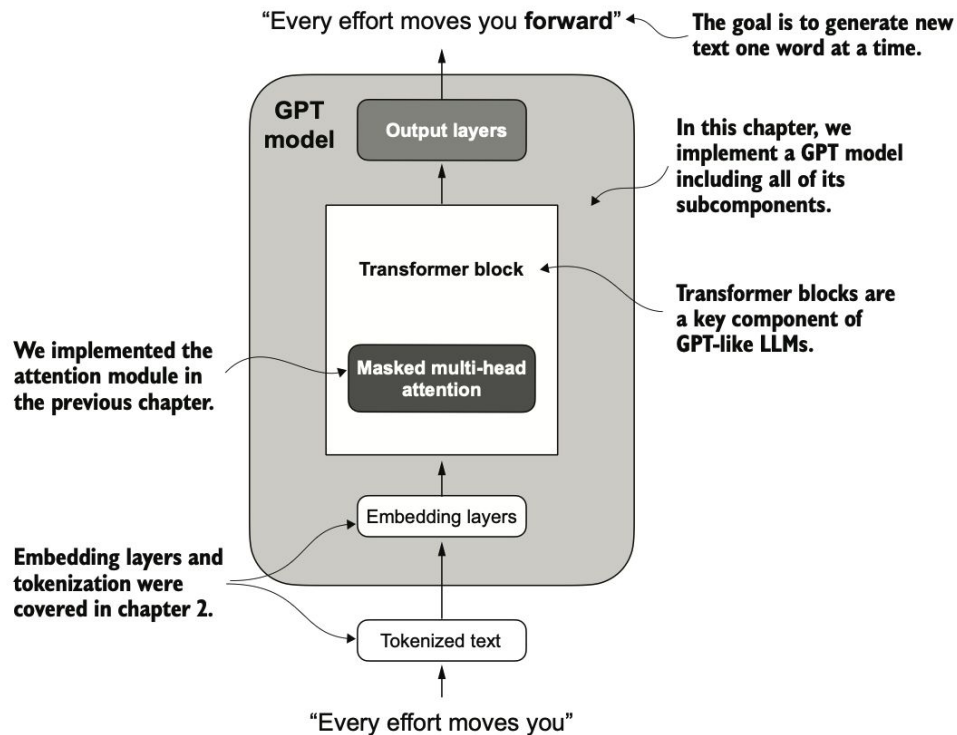
Chapter 4:

Implement GPT model from scratch

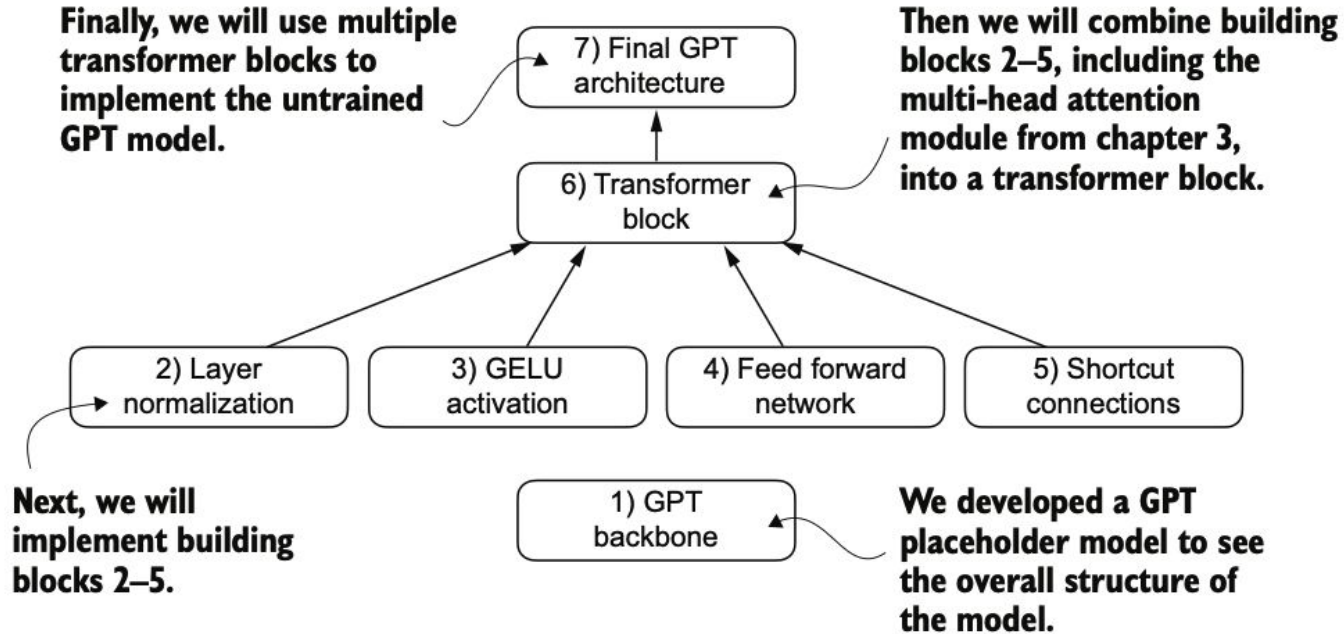
This chapter implements the LLM architecture

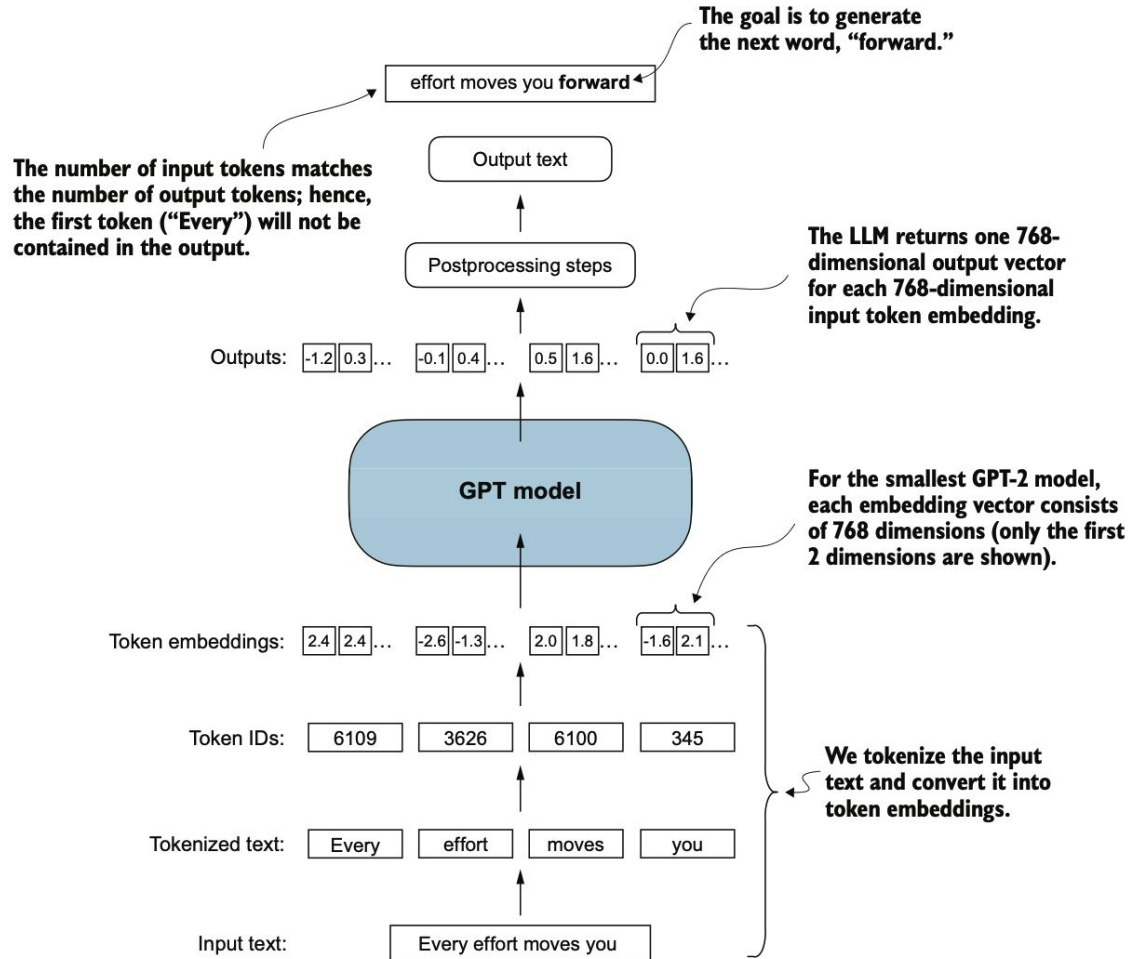


The transformer block

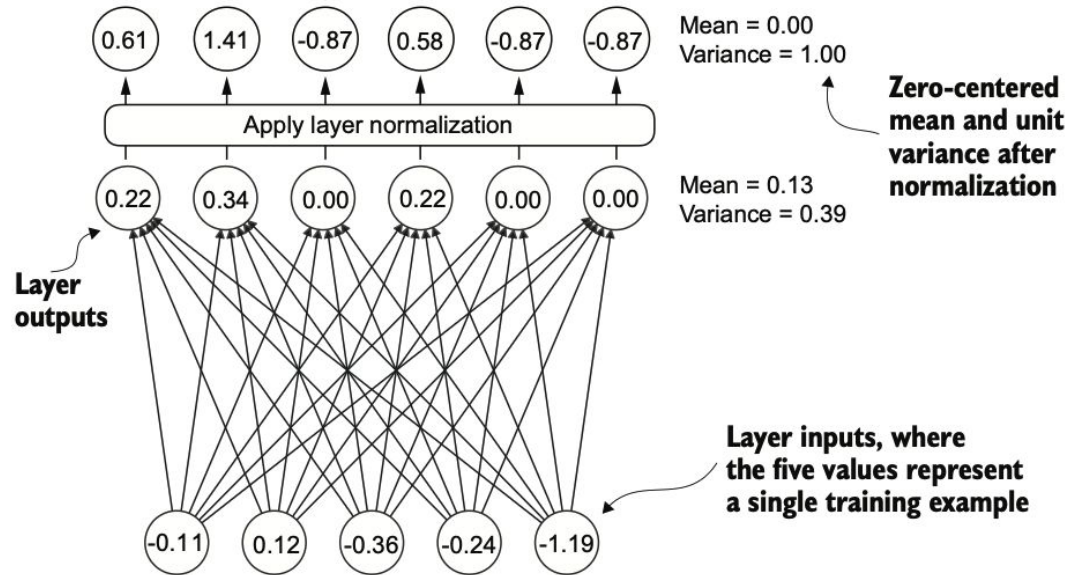


1) GPT Architecture



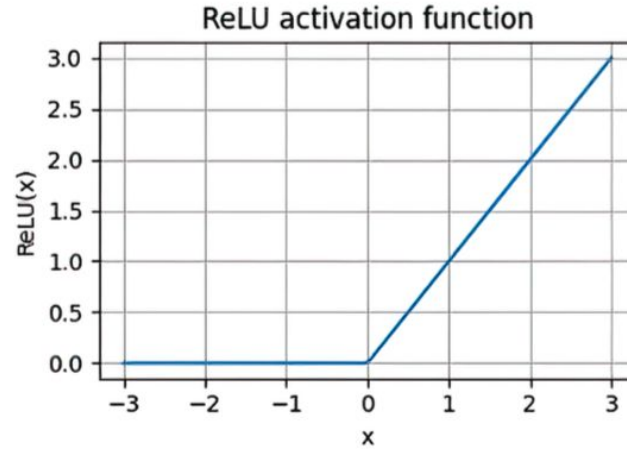
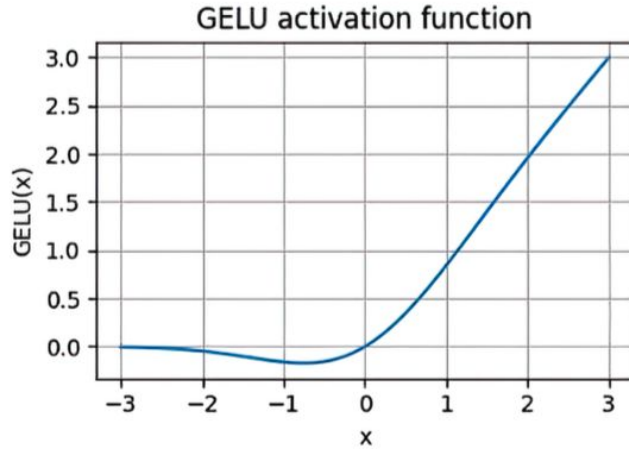


2) Layer Normalization

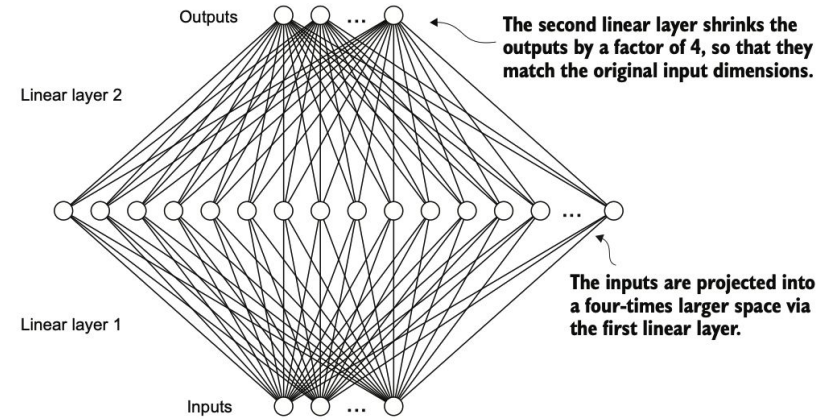
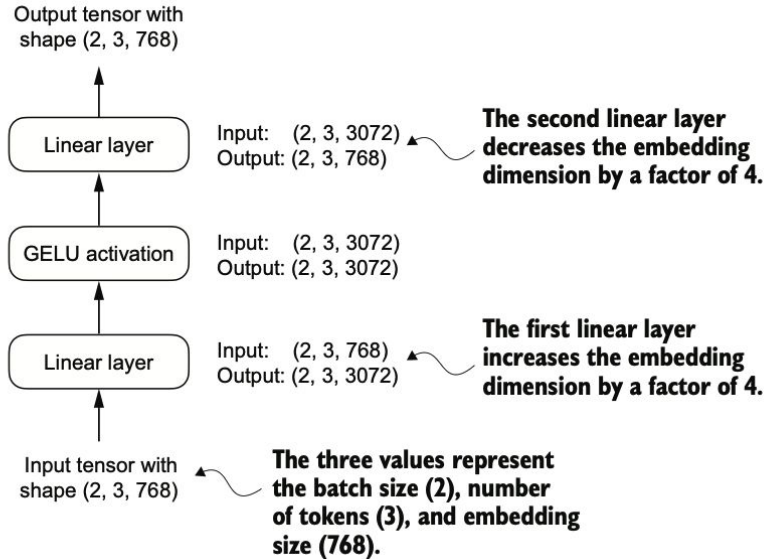


2) GELU Activations

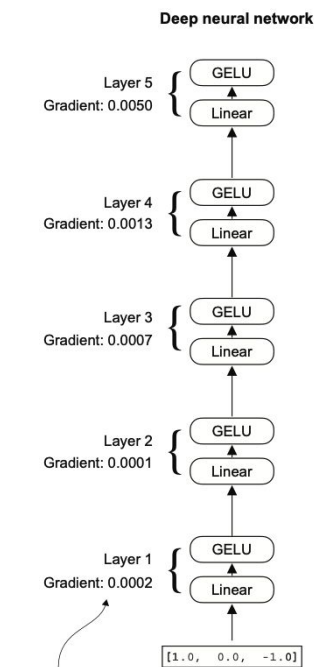
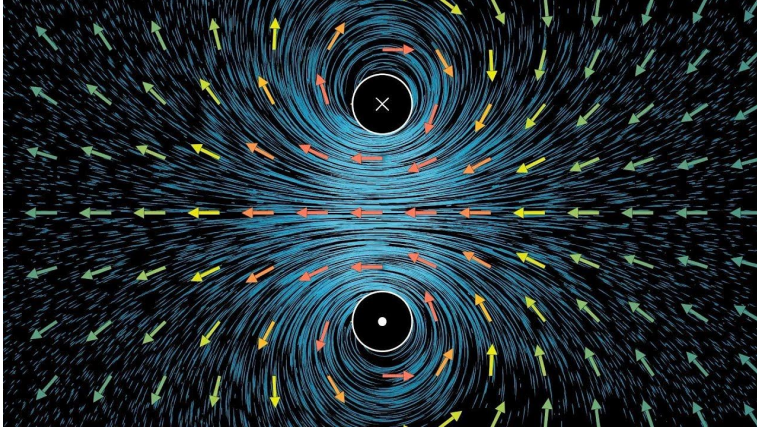
$$\text{GELU}(x) \approx 0.5 \cdot x \cdot \left(1 + \tanh \left[\sqrt{\frac{2}{\pi}} \cdot (x + 0.044715 \cdot x^3) \right] \right)$$



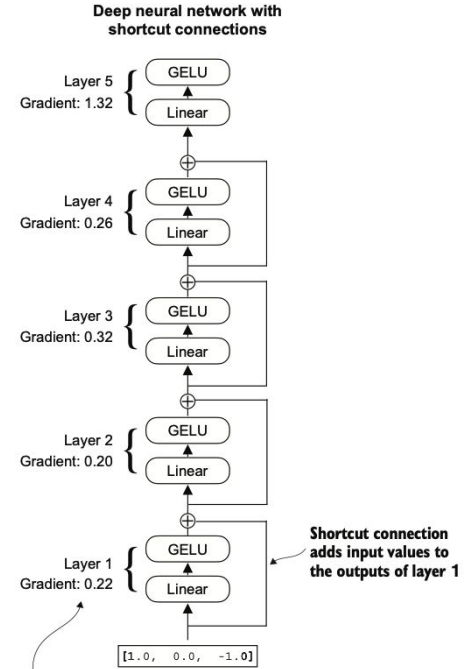
4) Feed Forward Network (Continue)



5) Shortcut Connections



In very deep networks, the gradient values in early layers become vanishingly small



The shortcut connections help with maintaining relatively large gradient values even in early layers

